Room plan: shuffle machines and loop names (permutations, part 2)

The plan for Wednesday, November 4, 2026 (James confirmed shuffle machines on November 2). Week 3, Shuffle machines: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose chair launch this plan uses. Unpiloted at the circle; tried at home.

Why this follows October 29

On October 29 the children counted the orders of 3 and 4 cards and built orders from swaps, with cycle notation planned but not reached. A shuffle machine is a whole order applied at once, and running it again and again splits the slots into loops. Cycle notation is just a name for those loops, so the children meet it as a way to write down something they walked through, not as a new rule.

For adults: the loops are the cycles; the number of turns until every cube is home is the least common multiple of the loop lengths (loops of 2 and 3 take 6 turns on 5 slots); a loop of k slots can be built from $k - 1$ swaps, which links back to last week's swaps and, for older children, to even and odd.

Prep checklist for Wednesday afternoon

Print single-sided, in colour, at 100% ("Actual size"), so the mat slots stay 1.2 inches. The guide's copy counts assume 4 children at K–1, 4 at 2–3 and 3 at 4–5; the counts below lean toward 2–3, since most of the room is there. Adjust at the door.

- K–1 packet, 10 pages: 4 copies, plus 4 extra copies of page 10 (the blank test mat) for the other tables
- Grades 2–3 packet, 11 pages: 8 copies
- Grades 4–5 packet, 8 pages: 4 copies
- Facilitator guide, 11 pages: one per adult (3)
- This room plan: one per adult (3)
- Opener sheet, 2 pages, cut in half: page 1 (the main puzzle, two per page) 5 copies, page 2 (easier on top, harder below) 3 copies
- Five animal signs: BEAR, FOX, OWL, RABBIT, TURTLE in big letters on five sheets of paper, for the sitters to hold
- Snap cubes from school: about 75, at least 14 each of green, blue, red, yellow and purple, plus a pink one per 2–3 child and a spare (any small pink object will do); one set per child in its own cup
- Masking tape and five coloured sheets for the chairs (green, blue, red, yellow, purple)
- Coloured pencils (three colours per child at 2–3), pencils, scrap paper
- Whiteboards and markers, one per child; one big whiteboard or easel by the chairs
- Sticky notes for the floor rounds
- Two decks of playing cards for 4–5, with one suit A to Q in order on top (the guide explains why)
- A timer everyone can see
- Ten-minute check the day before or that afternoon: the guide's three tests with real cubes (cubes fit the slots; K–1 Problem 3's bottom machine takes 6 turns; the 8-card perfect shuffle)
- At the room: set up the chairs and floor arrows before the children arrive (guide, "Set up before the children arrive")

Before the children arrive

- Opener half-sheets and pencils already on every table, so the first child in starts on the puzzle. The animal signs by the chairs.
- Five chairs with their coloured sheets and masking-tape arrows on the floor, exactly as in the guide (chairs 1 → 2 → 3 → 1, chairs 4 and 5 point to themselves).
- At each table: a cup of snap cubes per child (green, blue, red, yellow, purple, pink; about 75 in all), that

band's packet pages face down, coloured pencils, whiteboards.

- One timer everyone can see. A whiteboard by the chairs for loop names, and a pad of sticky notes.
- Printed early-finisher cards: "Draw a 5-slot machine that takes as many turns as you can" (2–3 and 4–5) and "Draw a machine where only one cube moves" (K–1, from its Problem 5; it can't be done).

Opener: who sat where? (8 minutes)

A soft logic puzzle on the tables as children arrive, in the spirit of year one's openers (the theater seats, Squirrel's party, the planets), then acted out on the same five chairs the launch uses. The answer becomes the launch's sitters, so the room goes straight from the puzzle into the first switch. An adult reads the clues aloud at any table that wants it.

The puzzle (most of the room). Five animals went to a show and sat in the five coloured chairs: 1 green, 2 blue, 3 red, 4 yellow, 5 purple. Write each animal's name in its chair.

1. Fox sat in the red chair.
2. Owl sat on an end chair.
3. Rabbit sat right next to Owl.
4. Bear sat somewhere to the left of Fox.
5. Turtle sat right next to Fox.

Answer: green Bear, blue Turtle, red Fox, yellow Rabbit, purple Owl. It is the only answer, and every clue is needed (checked by computer).

Hint ladder (give one step, then walk away):

1. "Which clue tells you exactly where one animal sat? Write that one first." (Fox, red.)
2. "Owl is on an end. Try Owl in green. Where does Rabbit go? Now where can Bear go?" (Rabbit blue, and Bear has no chair left of Fox: so Owl is not in green.)
3. "So Owl is in purple and Rabbit in yellow. Which clue places Turtle?" (Turtle blue, so Bear green.)

Easier (younger children, or anyone stuck at step 1). Three chairs: green, blue, red. Owl sat in the green chair. Fox did not sit next to Owl. Where did Bear sit? *Answer:* green Owl, blue Bear, red Fox.

Harder (older children, or anyone done in two minutes). The same five animals and chairs, but no clue names a chair: Owl sat on an end chair. Rabbit sat right next to Owl. Fox had exactly one animal between it and Owl. Bear sat somewhere to the left of Rabbit. Turtle did not sit on an end chair. *Answer:* the same seating as the main puzzle (Owl in green leaves Bear no room left of Rabbit).

Early finishers. "Cross out the clue *Turtle sat right next to Fox*. How many answers are there now? Write a new clue that brings it back to exactly one." (Two answers: Bear and Turtle can swap. "Bear sat on an end chair" fixes it, and so do others.)

On the floor (minute 6). Five children who solved it take the animal signs and sit where the answer says. The room checks each clue out loud, thumbs up or down. They stay seated: they are the launch's sitters, and the organizer hands each the cube of their chair's colour. A bridge question before the first switch: "After one switch, who will be in the red chair?" (Turtle, from blue.)

The hour

Time	Everyone	Jobs
0:00–0:08	**Opener: who sat where?** Puzzle on the tables as children arrive; at minute 6, five solvers act the answer out on the chairs.	Solvers check each other's sheets; adults give one hint step at a time.
0:08–0:11	**Launch at the chairs**, as in the guide: the five animal sitters stay, "switch" until everyone is home (3 switches). Practise the reset signal once.	Sitters walk; everyone else counts out loud.
0:11–0:13	**Loop names.** On the whiteboard, write the loop they just walked as (1 2 3) and say "1 goes to 2 goes to 3 goes back to 1". Chairs 4 and 5 are loops of one.	One child reads the loop name back.
0:13–0:28	**Tables, round 1**: Problem 1 in every band (run each machine until every cube is home). After each machine, the pair writes its loops on the whiteboard in loop names.	**Turner** moves the cubes; **Counter** keeps the tally and writes the loop names. Swap at every new machine.
0:28–0:33	**Back to the floor.** A pair from the 2–3 table turns one of their machines into instructions: each sitter gets a sticky note with the number of the chair they walk to (ignore the tape arrows). The room predicts the number of switches, then runs it.	New sitters; the pair is in charge of saying "switch".
0:33–0:48	**Tables, round 2**, by band. K–1: Problem 4 (draw arrows for 2, 3 and 4 turns; the partner checks with cubes). 2–3: Problem 2 (colour the loops, predict, then run), then Problem 3 (5 slots, 1 to 6 turns). 4–5: Problem 3, then "How many swaps build each of your machines?"	Same pairs, swapped jobs: the Predictor says a number before the Turner runs it.
0:48–0:55	**Floor finale: the slowest machine.** Each table brings its slowest 5-slot machine. Run the winner on the chairs with sticky notes (6 turns is the record for 5 slots).	Tables choose one sitter each.
0:55–1:00	Cubes back in cups by colour, pages collected.	Each pair has one colour to sort.

If it goes sideways

- A table finishes early: hand out its early-finisher card.
- A pair is stuck: the adult runs one turn with them and asks "Who is home now?", nothing more.
- The room gets loud: reset signal, then go straight to the next floor activity rather than talking over it.
- Jay and Rose have done the 4–5 pages at home. Make them floor captains at 0:28 and 0:48, and checkers at the 2–3 table who may say "try again" but not why.

How it went

(Add a dated line after the session.)